

# Arab Academy for Science, Technology & Maritime Transport (AAST)



## College of Computing & Information Technology

### CSE3101: Software Requirements and Specifications

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**Vision and Scope Document.**

## PARROT BOOKS SALES SYSTEM

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# **Vision and Scope Document**

**for**

## **<Parrot Books Sales System>**

**Version 1.0 approved**

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**<Arab Academy for Science, Technology, and Maritime Transport —  
College of Computing and Information Technology (CSE3101)>**

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## Revision History

| Name  | Date       | Reason For Changes                             | Version |
|-------|------------|------------------------------------------------|---------|
| O.D.A | 29/07/2026 | Initial draft based on Parrot Books case study | 1.0     |
|       |            |                                                |         |

# 1. Business Requirements

*This Vision and Scope Document defines the business rationale, scope, and stakeholder context for the Parrot Books Sales System, based on the “Current System Documentation” case study (manual Sales Order and Sales Ledger processes at Parrot Books Ltd).*

## 1.1. Background

*Parrot Books Ltd is a distributor of quality books serving a diverse and international client base, with 170 employees and annual revenue of EGP 14 million (2009), and a three-time winner of the Sloper Book Distributorship award. Its Sales Order and Sales Ledger processes are currently manual and screen-based: sales staff key orders into isolated screens, warehouse staff generate picking tickets and dispatch notes manually, and accounts staff reconcile cheques and remittance advices by hand, re-keying the same data at every stage. As the company's international order volume grows, this manual process has become a bottleneck, motivating the move to an integrated, web-based Sales System.*

## 1.2. Business Opportunity

*Competing book distributors increasingly offer direct online ordering with real-time stock visibility and order tracking, while automating requisitioning, picking, dispatch, invoicing, and statement generation internally. Parrot Books' manual process cannot match these turnaround times and does not scale efficiently as international order volume grows.*

*An integrated web-based Sales System removes repeated re-entry of the same order data across the Sales Order and Sales Ledger functions, gives management real-time visibility of stock, orders, and receivables, and lays the groundwork for a future customer self-service ordering portal — something the current manual system cannot offer.*

## 1.3. Business Objectives

- Reduce end-to-end order-to-dispatch cycle time by at least 40% within the first year of deployment.
- Reduce order data-entry and pricing errors caused by manual re-keying by at least 90%.
- Provide management with real-time visibility of stock levels, order status, and outstanding customer balances.
- Enable customers to place and track orders online, reducing dependence on paper-based order intake.
- Reduce the time to prepare monthly customer statements from days to hours through automation.
- Support the company's continued international growth without a proportional increase in administrative headcount.

## 1.4. Success Metrics

- Average order acknowledgement turnaround time (target: same business day).
- Percentage of orders requiring manual correction after entry (target: under 2%).
- Days Sales Outstanding (DSO) reduction versus current manual-process baseline.
- Customer satisfaction score for order placement and tracking (post-launch survey).
- System uptime and successful order-processing rate once the system is live (target: 99.5% availability).
- Reduction in warehouse picking errors and partial-shipment incidents.

## 1.5. Vision Statement

*For Parrot Books Ltd, staff, and international customers who need fast, accurate, and transparent book ordering and fulfillment, the Parrot Books Sales System is a web-based sales and order management platform that replaces disconnected manual and screen-based processes with a single, integrated system for order capture, stock and requisitioning, warehouse fulfillment, invoicing, and payment reconciliation. Unlike the current manual process, the new system gives every stakeholder — from the customer to the credit control clerk — a real-time, accurate view of an order's status from placement to payment, while preserving the company's reputation for excellence, customer service, and quality across all continents.*

## 1.6. Business Risks

| Risk                                                                                    | Severity | Mitigation                                                                                                          |
|-----------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------|
| Staff resistance to moving away from familiar manual/screen forms                       | Medium   | Phased rollout, hands-on training, and a UI that mirrors familiar document layouts (order, picking ticket, invoice) |
| Data migration errors from existing paper/screen records (customers, books, publishers) | High     | Dedicated data-cleansing and migration phase with reconciliation reports before cut-over                            |
| Integration gaps with the still-manual Purchase Order / Publisher process               | Medium   | Define clear data hand-off points (requisition, GRN) and keep Phase 1 interfaces file/paper-compatible              |
| Project schedule slippage given fixed academic/business deadline                        | Medium   | Prioritized MVP scope (Section 2.2) with subsequent releases deferred (Section 2.3)                                 |
| Security of online customer ordering and payment-related data                           | High     | Role-based access control, encrypted transport (HTTPS/TLS), and no storage of raw payment card data in Phase 1      |

## 1.7. Business Assumptions and Dependencies

- The company will provide the existing Books, Customers, and Publishers reference data for migration into the new system.
- Publishers will continue to receive Purchase Orders and issue Goods Received Notes through the existing paper/manual channel in the initial release; no publisher-side system integration is assumed.
- Bank deposit and cheque processing remain manual in Phase 1; the system produces the deposit-slip data but does not integrate directly with a banking API.
- Staff will have access to standard workstations with a modern web browser and internet/intranet connectivity.
- The Vision and Scope defined here is a subset of the long-term goal of a fully online, customer-facing ordering platform (see Section 2.3, Scope of Subsequent Releases).

## 2. Scope and Limitations

*This section defines what the Parrot Books Sales System will and will not deliver in its initial release, so that expectations among sales, warehouse, accounts, management, and customer stakeholders are aligned, and so that proposed changes can be evaluated against a clear baseline.*

### 2.1. Major Features

- 1. Sales Order Management — order capture, validation, and acknowledgement.
- 2. Stock & Purchasing — stock-level checking and purchase requisition/order generation.
- 3. Warehouse & Fulfillment — picking ticket and dispatch note generation.
- 4. Sales Ledger & Invoicing — invoice generation and ledger updates.
- 5. Payments & Cash Receipts — remittance matching and bank deposit preparation.
- 6. Reporting & Administration — catalogue/customer maintenance, reporting, and user access control.

*Unlike the current manual system, every feature above operates on a single shared order record, eliminating the repeated re-keying described in the case study.*

### 2.2. Scope of Initial Release

The initial release (Release 1.0) delivers a web-based application, used by sales, warehouse, shipping, and accounts staff (with a customer-facing order-placement view), covering:

- Digital capture and validation of sales orders against the book catalogue and pricing data.
- Real-time stock-on-hand checking with automatic flagging of shortfalls.
- Generation of purchase requisitions grouped by publisher when stock is insufficient.
- Generation of sales order acknowledgements, picking tickets, and dispatch notes from a single order record (eliminating re-keying).
- Generation of sales invoices and monthly customer statements from the sales ledger.
- Recording of customer payments (cheque/remittance matching) and preparation of bank deposit records.
- Role-based dashboards and basic operational reports for management.

### 2.3. Scope of Subsequent Releases

- Release 2.0: Full customer self-service portal with order tracking and order history.
- Release 2.0: Electronic payment gateway integration (card/online payment) in addition to cheque-based payment recording.
- Release 3.0: Publisher-facing portal for electronic purchase orders and goods-received confirmation.
- Release 3.0: Business-intelligence dashboards, demand forecasting, and multi-currency support for international customers.

## 2.4. Limitations and Exclusions

- The system is not a full general-ledger/accounting package; it produces the data needed to update the nominal and sales ledgers but does not replace the company's broader financial accounting system.
- Direct bank API integration and automated cheque image processing are excluded from Release 1.0.
- Warehouse automation (e.g., robotic picking, barcode conveyor integration) is out of scope; the system supports manual picking workflows.
- Human-resources, payroll, and supplier-contract-management functions are not part of this system.

## 3. Business Context

*This section profiles the Parrot Books Sales System's stakeholders, the project's priorities among schedule, features, quality, staffing, and cost, and the considerations for deploying the system into the company's operating environment.*

### 3.1. Stakeholder Profiles

*Stakeholders are individuals, groups, or organizations actively involved in the Parrot Books Sales System project, affected by its outcome, or able to influence its outcome. The stakeholder profiles below identify the customers and other stakeholders for this product and state their major interests in the product. For each stakeholder category, the profile below includes the major value or benefits they will receive from the product, their likely attitudes toward it, major interests, and any known constraints that must be accommodated:*

- *improved productivity*
- *reduced rework*
- *cost savings*
- *streamlined business processes*
- *automation of previously manual tasks*
- *ability to perform entirely new tasks or functions*
- *conformance to current standards or regulations*
- *improved usability or reduced frustration level compared to current applications*

*The stakeholder profile below is specific to the Parrot Books Sales System:*

| Stakeholder                     | Major Value                                                                               | Attitudes                                                                       | Major Interests                                           | Constraints                                                  |
|---------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Company Executives / Management | Faster order cycle, lower operating cost, real-time visibility into sales and receivables | Supportive; view the system as necessary to sustain international growth        | Reporting, scalability, return on investment              | Fixed project budget and academic/business delivery deadline |
| Sales Staff                     | Faster, less error-prone order entry; single source of order data                         | Cautiously receptive; concerned about relearning workflows                      | Ease of use, minimal re-keying, clear validation feedback | Must be usable with minimal retraining time                  |
| Warehouse & Shipping Staff      | Clear, accurate picking tickets and dispatch notes generated automatically                | Receptive if the interface is at least as simple as the current printed tickets | Reliable stock data, simple screen layout                 | Workstations on the warehouse floor may be shared/basic      |
| Credit Control / Accounts Staff | Automated invoicing, statements, and payment matching                                     | Positive; reduces manual reconciliation effort                                  | Accuracy of ledger updates, audit trail                   | Existing nominal ledger data formats must be respected       |
| Cashier                         | Simplified bank deposit preparation                                                       | Neutral to positive                                                             | Accurate cheque/remittance matching                       | Manual bank submission process retained in Phase 1           |
| Customers                       | Faster acknowledgement, accurate invoices/statements, and (future) online ordering        | Positive; international clients expect digital ordering                         | Order transparency, accurate pricing, timely delivery     | Varying time zones and connectivity levels                   |
| Publishers                      | Continue receiving requisitions/purchase orders promptly                                  | Neutral; unaffected in Phase 1                                                  | Clear, correctly grouped purchase orders                  | Continue to operate via existing paper/manual channel        |



### 3.2. Project Priorities

*The priorities among the Parrot Books Sales System's requirements, schedule, and budget are summarized below, following the key-drivers / constraints / degrees-of-freedom approach of Creating a Software Engineering Culture by Karl E. Wieggers (Dorset House, 1996), applied to this project:*

| Dimension | Driver (objective)                                                              | Constraint (limits)                   | Degree of Freedom (allowable range)                              |
|-----------|---------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------|
| Schedule  | Release 1.0 available in time for the CSE3101 project deadline of 7 August 2026 | Fixed academic deadline               | N/A                                                              |
| Features  | Core sales-order-to-payment workflow must be functional end to end              | N/A                                   | 70–80% of high-priority features must be included in Release 1.0 |
| Quality   | Requirements and diagrams must be internally consistent and traceable           | N/A                                   | Minor cosmetic/UI polish may be deferred to later iterations     |
| Staff     | Group of student team members acting as the delivery team                       | Limited team size and available hours | N/A                                                              |
| Cost      | Minimize infrastructure cost for the initial (academic) deployment              | Student/academic budget               | Cloud hosting free-tier or local deployment acceptable           |

### 3.3. Deployment Considerations

*The system is deployed as a browser-accessible web application, reachable by staff on the company intranet/internet and, for order placement, by customers anywhere in the world — consistent with Parrot Books' international client base. Staff in different departments (sales, warehouse, shipping, accounts) require role-based access from workstations at the same physical site, so no multi-site latency issues are anticipated for Release 1.0.*

*Deployment activities include migrating existing Books, Customers, and Publishers reference data from the manual system; training staff on the new screens (deliberately mapped to the familiar paper documents in the case study); and running the manual and new systems in parallel for a short transition period to validate data accuracy before full cut-over.*